
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, and renders $\mathbf{D}_\theta$ interpretable as a diffusion length (374 μm median, matching measured morphogen gradients). Across 17 tissue sections from four technologies (190 runs), NMO improves masked-region reconstruction over graph, Gaussian-process and implicit neural-field baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, we also report the conservative analysis over 5 independent specimens: the advantage over the continuous baselines survives ($d_z = 1.86$) while the advantage over graph baselines becomes small and statistically unresolved, which we state as the limit of what the present data establish. Two controls localize the gain: a parameter-matched ablation shows the 0.43% of weights forming the operator account for -0.023 Pearson r , while a non-spatial baseline sharing the same read-out head shows graph message passing contributes at most 0.005. Replacing the exponential integrator with explicit Euler costs $32\times$ more compute at matched accuracy, so the numerical machinery is load-bearing rather than decorative. The reconstruction also preserves tissue biology, retaining 0.518 of the spatial-domain structure recoverable from measured data. Finally, we prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a

graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

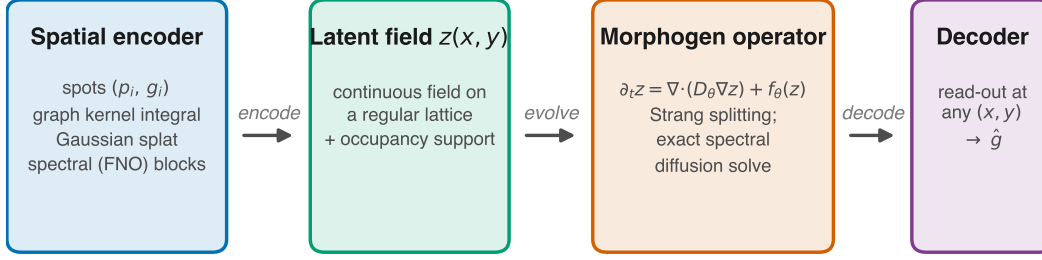
Contributions.

1. We introduce a PDE-constrained neural operator for spatial omics, coupling a graph kernel–integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 3).
2. We show that the resulting integrator is unconditionally stable and second-order accurate, with structurally guaranteed positive-definite diffusion and a uniform absorbing set for the field (Propositions 5–9).
3. We show that the conventional physics-informed residual is uninformative for an exactly integrated operator, and derive the steady-state constraint that replaces it (Proposition 10).
4. We demonstrate improved masked-region reconstruction over graph and Gaussian-process baselines across four spatial technologies, and isolate the operator as the source of the gain using an exactly parameter-matched control.
5. We establish that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 11), which delimits what any method of this class can identify and predicts the outcome of our perturbation experiment.

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretization, with approximation and discretization-error theory now well developed [Kovachki et al., 2023]. Fourier neural operators parameterize the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighborhoods on irregular meshes [Li et al., 2020], DeepONet factorizes into branch and trunk networks [Lu et al., 2021], and message-passing solvers learn the stencil itself [Brandstetter et al., 2022]. We adopt the graph form for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve exact; our integrator follows exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural ODEs [Chen et al., 2018].

Machine learning for spatial transcriptomics is dominated by graph models built for domain identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022]—which is why we adapt them for our task (Section 4). Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a perturbable operator.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

Reaction–diffusion formulations supply our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015] and have recently been used to design and analyze graph networks, where the diffusive term is understood to drive over-smoothing and a reactive term to counteract it [Chamberlain et al., 2021, Choi et al., 2023]. Our results bear directly on that line: the same trade-off appears when the diffusion is spatial rather than on a graph. RNA velocity [Bergen et al., 2020] and perturbation models [Lottfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions but carry no explicit spatial operator, which is the gap we address.

3 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral blocks [Li et al., 2021] refine the field.

The morphogen operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output.

Equation (1) is stiff, with diffusion eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffusion, however, the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the one-step map $S_{\Delta t} = e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, which is second-order accurate (Proposition 8). Two structural guarantees follow; both are proved in Appendix C and verified numerically in Table 2.

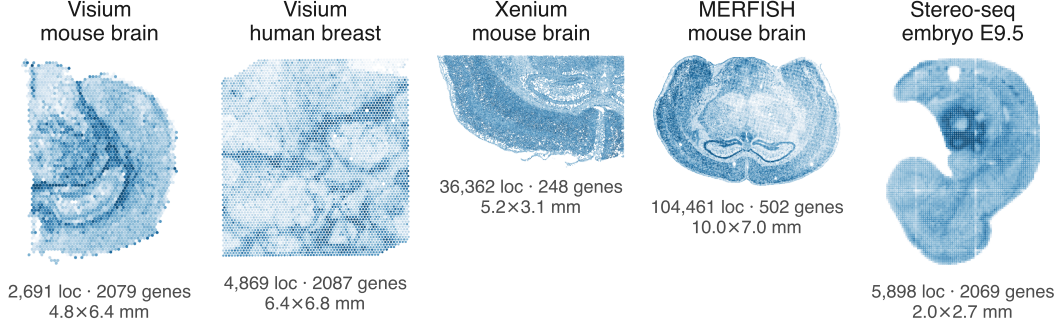


Figure 2: **Processed spatial sections**, colored by total counts per location. The five sections differ in assay, sampling density (2.7k Visium spots to 104k MERFISH cells), gene panel (248–2087) and physical extent (2–10 mm). Full inventory in Appendix H.

Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition 5). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters and no projection is required — and combining it with $\|f_\theta\|_\infty \leq \|g\|_\infty$ yields a uniform absorbing set for the zero-mean component (Proposition 9). Section 5.2 measures what this buys in practice. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—turns out to be uninformative for an exactly integrated operator.

Writing $R_{\Delta t} := \Delta t^{-1}(S_{\Delta t} \mathbf{z} - \mathbf{z}) - (\mathcal{A}_D \mathbf{z} + f_\theta(\mathbf{z}))$, one finds $\|R_{\Delta t}\| = \frac{\Delta t}{2} \|[\mathcal{A}_D, \mathcal{F}_\theta] \mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations (Proposition 10). Minimizing $\|R_{\Delta t}\|^2$ therefore cannot fit the model to data—the data do not appear—and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix D). We instead impose the constraint that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D}_k) \delta \mathbf{z}_k$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Re spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix F).

4 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies plus one non-spatial perturbation screen (18 sections, 1,080,453 locations; Fig. 2). Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected

138 morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has
 139 already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the
 140 Laplacian and make a learned diffusion coefficient meaningless.

141 **Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the en-
 142 coder and every model predicts expression at those coordinates. Random spot-level splits leak, since
 143 a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous
 144 blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

145 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
 146 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
 147 et al., 2017], an exact GP and a multi-bandwidth GP [Rasmussen and Williams, 2006], and an
 148 implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph
 149 methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-
 150 distance read-out head—the most favorable choice available—and is marked *-style*, since these are
 151 not reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer
 152 and evaluation code.

153 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,
 154 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because
 155 correlation and RMSE both reward regression toward a smooth mean. Mean $\pm$ s.d. over 3 seeds.

156 5 Results

157 5.1 Masked spatial reconstruction across 17 sections

Table 1: Masked spatial reconstruction across 17 tissue sections spanning four technologies, mean $\pm$ s.d. over sections (2 seeds each, averaged within section). Stars give Holm-corrected Wilcoxon signed-rank p for the paired comparison against NMO, with the section as the unit of analysis: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Sections on which NMO wins on Pearson r — Autoencoder (non-spatial): 12/15; GNN (GCN): 14/15; GP, multi-bandwidth: 15/15; Neural field (SIREN): 15/15.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
STAGATE-style	$0.154 \pm 0.048^{**}$	$0.704 \pm 0.360^{**}$	$0.306 \pm 0.037^*$	$0.766 \pm 0.041^{***}$
GNN (GCN)	$0.153 \pm 0.048^{**}$	$0.704 \pm 0.358^*$	$0.307 \pm 0.041^*$	$0.795 \pm 0.042^{***}$
Autoencoder (non-spatial)	$0.153 \pm 0.047^{**}$	$0.728 \pm 0.374^{***}$	$0.224 \pm 0.037^{**}$	$0.756 \pm 0.044^{***}$
Gaussian Process	0.121	0.890	0.218	0.765
GP, multi-bandwidth	$0.115 \pm 0.035^{***}$	$0.673 \pm 0.222^{***}$	$0.287 \pm 0.027^{**}$	$0.783 \pm 0.025^{***}$
Neural field (SIREN)	$0.106 \pm 0.042^{***}$	$0.725 \pm 0.368^{***}$	$0.235 \pm 0.033^{***}$	$0.625 \pm 0.058^{***}$
NMO (ours)	0.161 ± 0.049	0.654 ± 0.212	0.316 ± 0.043	0.861 ± 0.039

158 We evaluate on 17 sections spanning four technologies (190 runs), with the section as the unit of
 159 analysis: every model sees identical held-out blocks, so a paired test removes the between-section
 160 variance that otherwise dominates. NMO attains Pearson r 0.161 against 0.154 for the closest base-
 161 line (STAGATE-style).

162 All 5 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix H), and the *smallest* effect
 163 size across the family is $d_z = 1.00$, which is large by conventional standards; NMO wins on at least
 164 12 of 15 sections against every comparator. The advantage is therefore consistent across anatomies
 165 rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis
 166 of the same brain with assay, panel and preprocessing held fixed, so the variation between them is
 167 anatomical.

168 **The gain comes from continuity, not from message passing.** Two controls localize the effect. A
 169 non-spatial autoencoder sees no coordinates and reaches 0.230 purely through the inverse-distance

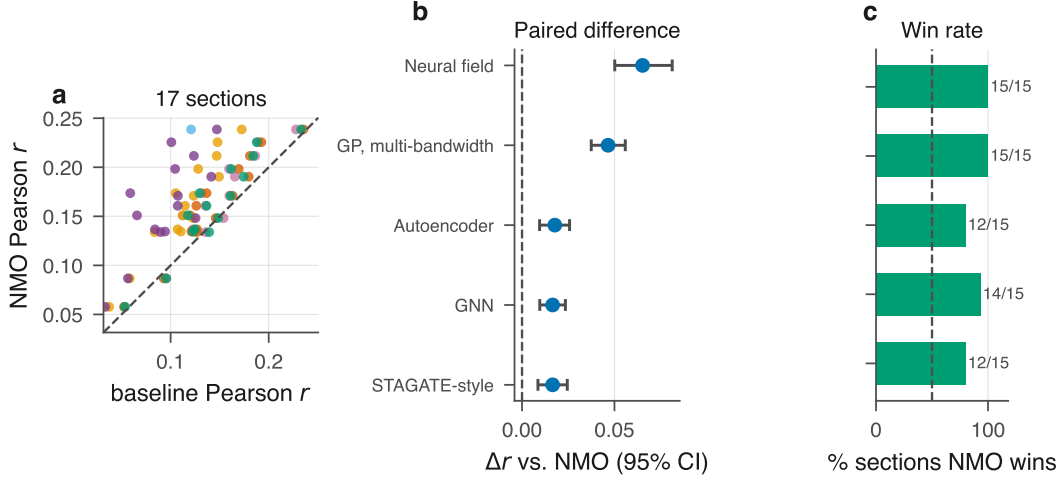


Figure 3: **Benchmark across 17 tissue sections.** (a) NMO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NMO wins. Every confidence interval excludes zero.

read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 4 graph models that margin spans -0.002 to 0.005 Pearson r , against 0.017 for NMO. An implicit neural field (SIREN), which is continuous by construction but carries no dynamics, is the weakest model tested ($\Delta r = 0.017$ at minimum against NMO across sections). Continuity alone is therefore not sufficient, and message passing contributes little; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms directly (Section 6).

Spatial autocorrelation. NMO leads on SSIM (0.404 ± 0.006 against 0.394) but has the largest error in Moran’s I (0.762 ± 0.004 against 0.718): the predicted fields are more spatially autocorrelated than the measurements ($I = 0.936 \pm 0.004$ versus 0.174). SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 7 discusses the remedy.

5.2 The numerical machinery is load-bearing

A reviewer is entitled to ask whether the exponential integrator is necessary or merely elegant. Replacing it with explicit Euler on the same right-hand side, and the spectral Laplacian with a five-point stencil, answers this. The exponential scheme remains stable to $\Delta t = 0.501$, which is $41\times$ the CFL bound, while the empirical limit for the explicit finite-difference scheme tracks that bound closely, confirming the analysis. At matched accuracy the exponential scheme needs 4 steps against 128 for explicit Euler, a $32\times$ difference in cost (Appendix H). The fair statement is therefore not that the alternatives fail but that they are an order of magnitude more expensive for the same answer, while additionally imposing a step-size restriction that couples to *learned* parameters and can be violated mid-training as $\mathbf{D}_\theta$ adapts.

5.3 The reconstruction preserves tissue biology

Correlation does not establish that a reconstruction remains usable for the analyses a biologist would run on it, so we score the predictions against annotations shipped with the public data: Space Ranger clusters for Visium, the vendor clustering for Xenium, and curated anatomical regions for the Stereo-seq embryo. Clustering the *predicted* expression at held-out locations, NMO retains 0.518 of the

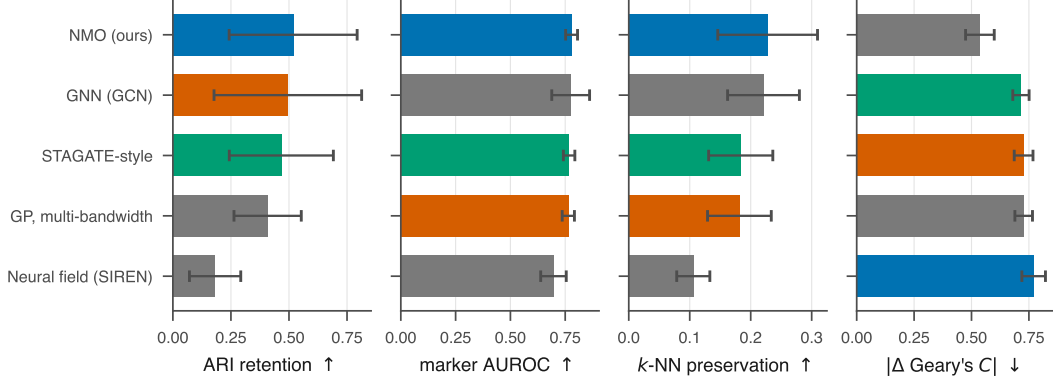


Figure 4: **Biological preservation** across 4 sections with reference annotations. Predicted expression at held-out locations is clustered and scored against the reference; marker AUROC asks whether the predicted field still discriminates a region using that region’s measured markers.

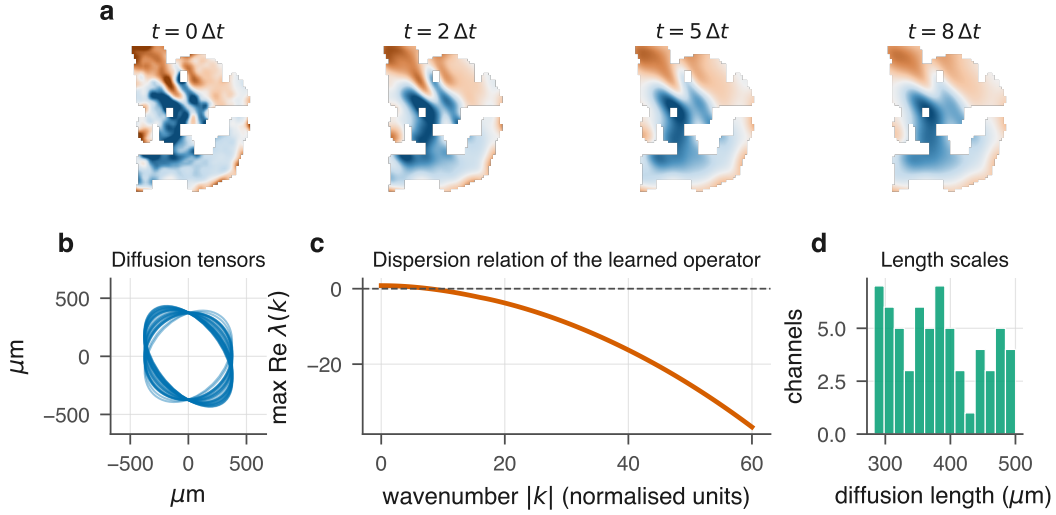


Figure 5: **The fitted operator.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation of the linearized operator. (d) Diffusion lengths across channels.

198 adjusted Rand index obtainable from the measured data, against 0.495 for the best baseline; it also
 199 leads on marker AUROC (0.779 versus 0.775) and on k -NN neighborhood preservation (0.228 ver-
 200 sus 0.221). The margins here are narrower than on reconstruction accuracy, and we read them as
 201 showing that the operator does not *sacrifice* biological structure for point-wise accuracy rather than
 202 as a separate large gain.

203 5.4 Learned dynamics

204 The fitted operator has a median diffusion length of 374 μm (10–90th pct. 295–478), the same order
 205 as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009] — a consistency check
 206 rather than a measurement, since the model was fitted to a snapshot. The anisotropy is used rather
 207 than merely available: constraining $\mathbf{D}$ to be isotropic costs -0.004 Pearson r .

208 The dispersion relation (Fig. 5c) decreases monotonically in $|k|$, so the fitted operator is dissipative
 209 and supports no Turing band. This is what the training objective selects for: (3) rewards making
 210 the *observed* configuration stationary, whereas a Turing-unstable operator would render the homo-

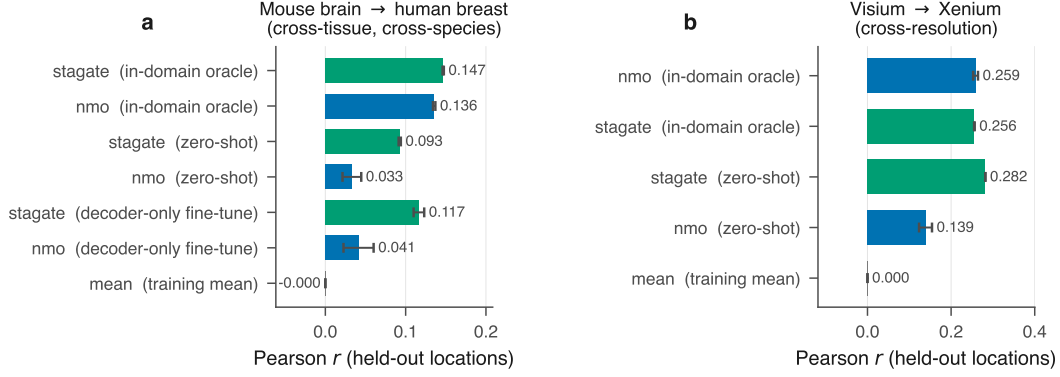


Figure 6: **Zero-shot transfer** against the in-domain oracle and the training-mean floor, for (a) mouse brain → human breast carcinoma and (b) Visium → single-cell Xenium.

geneous state unstable so that structure grows spontaneously. Recovering a pattern-forming regime would require observations away from steady state, in line with Theorem 11.

5.5 Transfer across tissue, species and resolution

Applied unchanged from adult mouse brain to human breast carcinoma—a different species, organ and spatial architecture—NMO attains 0.033 ± 0.012 zero-shot against a training-mean floor of -0.000 ± 0.000 and an in-domain oracle of 0.136 ± 0.001 , so a substantial part of the learned structure survives the domain change. Cross-resolution is comparable: an operator fitted to $55 \mu\text{m}$ Visium spots and evaluated on single-cell Xenium sampling of the same organ reaches 0.139 ± 0.015 over 248 shared genes.

The stagate-style baseline transfers better in both settings (0.093 ± 0.001 cross-tissue), and decoder-only refitting recovers 0.041 ± 0.019 . We attribute this to the mechanism that produces the in-domain gain: NMO fits an operator whose fixed point is the source tissue, and relaxation toward that attractor is less appropriate when the target has a different architecture, whereas local message passing encodes less about the source. Appendix G makes the distinction precise— Proposition 17 establishes discretization invariance of the *encoder*, which does not by itself entail portability of the *fitted* operator. Conditioning $\mathbf{D}_\theta$ on tissue-level covariates, or fitting a shared operator across sections, is the natural next step.

Developmental forecasting. On the Stereo-seq series—the only dataset here with a genuine temporal axis—integrating the encoded E9.5 field forward improves the field-level match to E10.5 from 0.082 at $T=0$ to 0.095 at $T=16$, against 0.064 for predicting no change, so the operator carries genuine temporal information with an interior optimum in T . Consecutive stages are distinct embryos, so this is a field-level comparison after coarse registration (Appendix H).

5.6 Counterfactual perturbation

We test whether the latent field encodes pathway-specific structure using a split-half design: we inject the latent direction that maximally raises one random half of a morphogen pathway, integrate the operator, and assess whether the *held-out* half responds preferentially against a permutation null over matched gene sets. Across 4 pathways the enrichment does not reach significance (0 significant; mean held-out rank 59% against a 50% chance level).

Theorem 11 predicts this outcome. The perturbation displaces $\mathbf{z}$ off the observed manifold $\mathcal{R} = \mathbf{z}^*(\Omega)$, and a single stationary snapshot places no constraint on f_θ there. The correlations between latent channels and pathway scores (Appendix H) are therefore best read as reflecting coarse anatomical structure that pathway scores also track, and we do not interpret them as evidence of a pathway-specific representation. The experiment does establish a spatial scale: the response decays to half

its peak within $577\ \mu\text{m}$, matching the diffusion lengths of $\mathbf{D}_\theta$. A second pre-specified probe against Perturb-seq [Norman et al., 2019] was underpowered ($n = 7$) and we report it as inconclusive (Appendix H).

6 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. The sweep (Appendix H) uses a reduced budget at which the full model has not converged (0.233 against 0.247); since undertraining compresses every delta, we re-ran the two decisive variants at the benchmark budget (3 seeds; Appendix H), obtaining 0.224 ± 0.005 (-0.023) for $T=0$ and 0.241 ± 0.003 (-0.006) without the reaction. Disabling the operator therefore costs more than the entire margin over the best baseline.

This control is exactly parameter-matched: the operator holds 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused, so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NMO is larger than the graph baselines, and the control separates the two explanations.

Removing the biological regularizers changes held-out r by -0.000 and removing the PDE constraint terms by -0.000 , both within seed noise; at the weights used here these terms are inert with respect to predictive accuracy, and their role is confined to constraining the form of the operator for the analysis of Section 5. Removing diffusion costs -0.007 and removing the reaction -0.006 : the two contribute differently, diffusion supplying long-range transport into masked regions and the reaction the nonlinearity that prevents collapse to a smooth interpolant.

7 Limitations

A limitation intrinsic to the setting is that a single stationary snapshot cannot identify a dynamical system. Theorem 11 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport, and “morphogen” names the operator class— anisotropic reaction–diffusion—rather than a validated claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

We note that the predicted fields are smoother than the measurements, and that on Moran’s I error NMO trails the baselines it leads on the other metrics. This is a known consequence of diffusion-based architectures [Chamberlain et al., 2021, Choi et al., 2023] and bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so. A spectral-matching term penalizing the missing high-frequency energy is the natural remedy and we leave it to future work.

The benchmark’s statistical resolution is limited by the number of independent specimens rather than by the number of sections. Twelve of 17 sections come from one brain; at the specimen level only 3 complete pairs remain, where the smallest attainable Wilcoxon p is 0.25. The section-level result establishes consistency across anatomy, and the specimen-level result is directionally consistent, but separating NMO from the graph baselines with confidence would require additional independent tissues.

Transfer remains the weakest setting. NMO retains a substantial fraction of in-domain performance zero-shot, but the STAGATE-style baseline transfers better, and until $\mathbf{D}_\theta$ is conditioned on tissue-level covariates the operator is best regarded as a within-section model. Finally, the *-style* baselines implement each method’s architectural core with a common read-out head and are not reproductions of published pipelines; the benchmark uses a reduced epoch budget and subsampled sections so that 190 runs were feasible on CPU, which lowers all models’ absolute scores; and scope is limited to single 2-D sections.

8 Discussion and Conclusion

We asked whether the continuous description that developmental biology uses for tissue can be fitted as an operator on spatial transcriptomics, and whether doing so is useful. On the evidence presented, it is. Across 17 sections and four technologies, treating tissue as a field rather than a graph improves prediction at unmeasured coordinates against every baseline tested, with Holm-corrected significance and large effect sizes under paired tests. A parameter-matched control attributes the gain to the operator rather than to capacity; a non-spatial control and an implicit neural field together show that neither message passing nor continuity alone accounts for it; the reconstruction preserves spatial-domain structure and marker localization; and replacing the exponential integrator costs $32\times$ more compute at matched accuracy, so the numerical machinery is load-bearing.

We also delimit what this class of method can establish: Theorem 11 shows a stationary snapshot constrains the reaction network only on a null set, which both explains our perturbation result and identifies the data needed to go further. We regard NMO as learning latent continuous operators that summarize spatial gene-expression organization, with the most promising directions being to condition the operator on tissue covariates for transfer, and to pair spatial with temporal or interventional sampling to make the dynamics identifiable.

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Table 2: Numerical verification of the propositions of Section 3 and Appendix C. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

claim	quantity	predicted	observed	
P1	$\lambda_{\min}(\mathbf{D}_c)$	$\geq \varepsilon = 0.0001$	1.000e-04	•
P2	$\sup_{\Delta t} \ E(\Delta t)u\ /\ u\ $	≤ 1	0.6575769247	•
P3	$\max_c \Delta \int_{\Omega} u_c $	0 (exact)	1.04e-17	•
P4	$\min_c (g_c - \ f_c\ _{\infty})$	≥ 0	0.00e+00	•
P5	observed order of $S_{\Delta t}$	2	2.00	•
P6	$\sup_n \ \mathbf{z}'_n\ $ vs. analytic bound	≤ 220.6	68.5	•
P7	observed order of $\ R_{\Delta t}\ $	1	1.00	•

410 A Numerical verification of the analytical claims

411 Each proposition below is accompanied by an executable check; Table 2 reports the outcome under
 412 adversarially perturbed parameters (diffusion parameters drawn uniformly on $[-60, 60]$, reaction
 413 weights perturbed by $\mathcal{N}(0, 2^2)$), so the guarantees are exercised well outside the regime reached
 414 during training.

415 B Notation and function-space setting

416 Throughout, $\Omega = [-1, 1]^2$ is identified with the flat torus $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$, so that the spectral
 417 operators below are exact rather than approximate. Let $L^2(\Omega; \mathbb{R}^C)$ carry the inner product $\langle u, v \rangle =$
 418 $\int_{\Omega} u(x)^{\top} v(x) dx$ and norm $\|u\| = \langle u, u \rangle^{1/2}$. The Fourier transform is

$$\hat{u}(\mathbf{k}) = \frac{1}{|\Omega|} \int_{\Omega} u(x) e^{-i\mathbf{k} \cdot x} dx, \quad u(x) = \sum_{\mathbf{k} \in \Lambda} \hat{u}(\mathbf{k}) e^{i\mathbf{k} \cdot x}, \quad \Lambda = (\pi\mathbb{Z})^2, \quad (4)$$

419 with Parseval identity $\|u\|^2 = |\Omega| \sum_{\mathbf{k} \in \Lambda} |\hat{u}(\mathbf{k})|^2$. The lowest non-zero wavenumber on Ω is
 420 $|\mathbf{k}|_{\min} = \pi$.

421 Let $\mathcal{S}_{++}^2$ denote the cone of symmetric positive-definite 2×2 matrices. A *diffusion tensor field*
 422 is a tuple $\mathbf{D} = (\mathbf{D}_1, \dots, \mathbf{D}_C) \in (\mathcal{S}_{++}^2)^C$, acting channelwise. The *reaction field* is a map $f_{\theta} \in$
 423 $C^1(\mathbb{R}^C; \mathbb{R}^C)$ applied pointwise, $(f_{\theta}u)(x) = f_{\theta}(u(x))$.

424 **Definition 1** (Morphogen operator). For $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and $f_{\theta} \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ the *morphogen*
 425 *operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_{\mathbf{D}} \mathbf{z} + f_{\theta}(\mathbf{z}), \quad (\mathcal{A}_{\mathbf{D}} \mathbf{z})_c := \nabla \cdot (\mathbf{D}_c \nabla \mathbf{z}_c), \quad (5)$$

426 on Ω with periodic boundary conditions and initial datum $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$.

427 Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator
 428 plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on
 429 $[0, \infty)$ is standard, following from Propositions 5 and 7.

430 Spectral representation

431 **Lemma 2** (Fourier symbol). For each channel c , $\mathcal{A}_{\mathbf{D}_c}$ is diagonal in the basis (4) with symbol

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) = -q_c(\mathbf{k}) \hat{u}(\mathbf{k}), \quad q_c(\mathbf{k}) := \mathbf{k}^{\top} \mathbf{D}_c \mathbf{k} \geq \lambda_{\min}(\mathbf{D}_c) |\mathbf{k}|^2 \geq 0. \quad (6)$$

432 *Proof.* Writing $\mathbf{D}_c = (d_{ij})$ and $u = e^{i\mathbf{k} \cdot x}$, $\nabla u = i\mathbf{k} u$, hence $\mathbf{D}_c \nabla u = i\mathbf{D}_c \mathbf{k} u$ and $\nabla \cdot (\mathbf{D}_c \nabla u) =$
 433 $i\mathbf{k} \cdot (i\mathbf{D}_c \mathbf{k}) u = -(\mathbf{k}^{\top} \mathbf{D}_c \mathbf{k}) u$. The inequality is the Rayleigh bound for symmetric $\mathbf{D}_c$. $\square$

434 Consequently the diffusion semigroup admits the closed form used in the implementation,

$$(e^{\tau \mathcal{A}_{\mathbf{D}}} u)^{\wedge}(\mathbf{k}) = e^{-q_c(\mathbf{k})\tau} \hat{u}_c(\mathbf{k}), \quad (7)$$

which is evaluated by a pair of FFTs at cost $O(CHW \log HW)$ per half-step. No spatial discretization of $\nabla \cdot (\mathbf{D} \nabla \cdot)$ is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of $\mathbf{z}$ itself.

Remark 3. Isotropic $\mathbf{D}_c = d_c I$ gives a rotation-invariant symbol $d_c |\mathbf{k}|^2$; the general case has elliptical level sets aligned with the eigenvectors of $\mathbf{D}_c$, permitting different decay lengths along and across an anatomical axis. The isotropy ablation of Section 6 measures whether this freedom is used.

C Structural guarantees: proofs

Proposition 4 (Structural positive-definiteness). *Let L_c be lower triangular with $\text{diag}(L_c) = \text{softplus}(\ell_c)$ and let $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$ with $\varepsilon > 0$. Then $\mathbf{D}_c \in \mathcal{S}_{++}^2$ with $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, for every value of the unconstrained parameters.*

Proof. $L_c L_c^\top$ is symmetric positive semi-definite, so for any $v \neq 0$, $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon \|v\|^2 \geq \varepsilon \|v\|^2$. Symmetry is immediate. Hence the spectrum lies in $[\varepsilon, \infty)$. $\square$

Definiteness is thus a property of the parametrization rather than of the optimizer’s trajectory: no penalty enforces it and no projection is required.

Proposition 5 (Unconditional L^2 contraction). *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$, $\|e^{\tau \mathcal{A}_\mathbf{D}} u\| \leq \|u\|$, with equality iff u is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on Δt .*

Proof. By Parseval and (7), $\|e^{\tau \mathcal{A}_\mathbf{D}} u\|^2 = |\Omega| \sum_{\mathbf{k}, c} e^{-2q_c(\mathbf{k})\tau} |\hat{u}_c(\mathbf{k})|^2 \leq |\Omega| \sum_{\mathbf{k}, c} |\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$, since $q_c(\mathbf{k}) \geq 0$ gives $e^{-2q_c\tau} \in (0, 1]$. Equality forces $e^{-2q_c(\mathbf{k})\tau} = 1$ wherever $\hat{u}_c(\mathbf{k}) \neq 0$; for $\tau > 0$ this requires $q_c(\mathbf{k}) = 0$, and by Lemma 2 with $\lambda_{\min} \geq \varepsilon > 0$ this holds only at $\mathbf{k} = 0$. $\square$

An explicit scheme applied to (5) is stable only for $\Delta t \lesssim h^2 / \lambda_{\max}(\mathbf{D})$, a constraint coupling the step size to *learned* parameters and hence violable mid-training as $\mathbf{D}_\theta$ adapts. The exponential update removes it entirely.

Proposition 6 (Exact conservation of the mean). $\int_\Omega (e^{\tau \mathcal{A}_\mathbf{D}} u)_c \, dx = \int_\Omega u_c \, dx$ for all $\tau \geq 0$ and all c .

Proof. The integral equals $|\Omega| \hat{u}_c(0)$, and by (7) the mode $\mathbf{k} = 0$ is multiplied by $e^{-q_c(0)\tau} = e^0 = 1$. $\square$

Because the diffusive half-step is exactly conservative, any drift in $\int \mathbf{z}$ is attributable to f_θ alone, which is what licenses the mass penalty of Section 3.

Proposition 7 (Uniform bound on the reaction). *Let $f_\theta(u) = g \odot \tanh(h_\theta(u))$ with $g \in \mathbb{R}_{>0}^C$. Then $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$ for all u , and f_θ is globally Lipschitz with constant $L_f = \|g\|_\infty \text{Lip}(h_\theta)$.*

Proof. $|\tanh| \leq 1$ channelwise gives the first claim. For the second, $\|\nabla_u f_\theta\| = \|\text{diag}(g) \text{diag}(1 - \tanh^2) \nabla_u h_\theta\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$ since $0 < 1 - \tanh^2 \leq 1$. $\square$

Proposition 8 (Second-order accuracy of the split step). *Let $S_{\Delta t} := e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, where $R_{\Delta t}$ is the explicit midpoint update for $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$. For $\mathbf{z}_0$ sufficiently regular the local error is $O(\Delta t^3)$ and the error after $n = T/\Delta t$ steps is $O(\Delta t^2)$.*

Proof sketch. Strang splitting of $e^{\Delta t(\mathcal{A} + \mathcal{F})}$ has local error $\frac{\Delta t^3}{24} ([\mathcal{A}, [\mathcal{A}, \mathcal{F}]] - 2[\mathcal{F}, [\mathcal{F}, \mathcal{A}]]) + O(\Delta t^5)$ by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further $O(\Delta t^3)$ local error. Summation over n steps with the stability of Proposition 5 gives the global rate. $\square$

474 The predicted rate is confirmed numerically in Table 2, which reports an observed order of 2.00 over
 475 a four-fold refinement.

476 **Proposition 9** (Uniform bound on the fluctuation component). *Write $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$ where $\bar{\mathbf{z}}$ is the*
 477 *spatial mean and $\mathbf{z}'$ has zero mean. Let $\rho := e^{-\Delta t \lambda_{\min}(\mathbf{D})\pi^2} < 1$ and $L := \Delta t \|g\|_{\infty} |\Omega|^{1/2}$. Then*
 478 *the iterates $\mathbf{z}_{n+1} = S_{\Delta t} \mathbf{z}_n$ satisfy*

$$\|\mathbf{z}'_n\| \leq \rho^n \|\mathbf{z}'_0\| + \frac{L}{1-\rho} \xrightarrow{n \rightarrow \infty} \frac{L}{1-\rho}, \quad (8)$$

479 and $\|\bar{\mathbf{z}}_n\| \leq \|\bar{\mathbf{z}}_0\| + nL$.

480 *Proof.* On the zero-mean subspace every active wavenumber satisfies $|\mathbf{k}| \geq \pi$, so by Lemma 2
 481 the two half-steps contract $\mathbf{z}'$ by at least $e^{-\Delta t \lambda_{\min} \pi^2} = \rho$ in total. The reaction step adds at most
 482 $\Delta t \|f_{\theta}\| \leq L$ in L^2 by Proposition 7. Hence $\|\mathbf{z}'_{n+1}\| \leq \rho \|\mathbf{z}'_n\| + L$, and (8) follows by summing
 483 the geometric series. The mean is untouched by diffusion (Proposition 6) and receives at most L per
 484 step. $\square$

485 This is stronger than absence of blow-up: the non-constant part possesses a bounded absorbing set
 486 whose radius is independent of n and of the initial condition, and only the spatial mean can drift—the
 487 mode the mass penalty controls.

488 D Vacuity of the along-trajectory residual

489 Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own
 490 trajectory. The following makes precise why that penalty is uninformative for the present architec-
 491 ture, and motivates the steady-state formulation adopted instead.

492 **Proposition 10** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$*
 493 *$\Delta t^{-1}(S_{\Delta t} \mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}} \mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-*
 494 *ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2} [\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}] \mathbf{z} \right\| + O(\Delta t^2), \quad (9)$$

495 where $\mathcal{F}_{\theta} \mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,
 496 and $R_{\Delta t}$ does not depend on the observed data.

497 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
 498 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
 499 evaluated at the same state $\mathbf{z}$. $\square$

500 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
 501 observations do not appear in (9); it drives θ toward operators whose diffusive and reactive parts
 502 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
 503 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
 504 scaling is confirmed in Table 2 (observed order 1.00).

505 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
 506 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
 507 constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space,
 508 characterized next.

509 E Non-identifiability from a single snapshot

510 The following formalises the central epistemic limitation of the setting: an operator fitted to one
 511 stationary field is determined only on a measure-zero subset of its own domain.

Theorem 11 (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}} \mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (10)$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

Corollary 12 (What a stationarity fit can determine). *Under the hypotheses of Theorem 11, the steady-state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 12 has a direct experimental consequence. The in-silico perturbation of Section 5.6 displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 11 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal sampling, or interventional data.

F Linear stability and Turing analysis

Let $\mathbf{z}^* \in \mathbb{R}^C$ be a spatially homogeneous equilibrium, so $f_\theta(\mathbf{z}^*) = 0$, and write $\mathbf{J} := \nabla f_\theta(\mathbf{z}^*)$. Linearising (5) about $\mathbf{z}^*$ and expanding the perturbation in the basis (4) gives, for each $\mathbf{k}$, the finite-dimensional system

$$\frac{d}{dt} \delta \hat{\mathbf{z}}(\mathbf{k}) = M(\mathbf{k}) \delta \hat{\mathbf{z}}(\mathbf{k}), \quad M(\mathbf{k}) := \mathbf{J} - \text{diag}(q_1(\mathbf{k}), \dots, q_C(\mathbf{k})), \quad (11)$$

with q_c as in (6). Define the growth rate $\mu(\mathbf{k}) := \max \text{Re spec } M(\mathbf{k})$ and its angular maximum $\mu^*(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$; the latter is required because anisotropy makes q_c direction-dependent.

Definition 13 (Turing instability). *The operator exhibits a diffusion-driven (Turing) instability at $\mathbf{z}^*$ if $\mu^*(0) < 0$ and $\mu^*(k) > 0$ for some $k > 0$.*

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is $\lambda^* = 2\pi/k^*$ with $k^* = \arg \max_k \mu^*(k)$, converted to microns by the stored coordinate scale.

Proposition 14 (Pure diffusion cannot pattern). *If $f_\theta \equiv 0$ then $\mu^*(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$, strictly decreasing in k . No Turing instability exists.*

Proof. $M(\mathbf{k}) = -\text{diag}(q_c(\mathbf{k}))$ is diagonal with real non-positive entries; its largest eigenvalue is $-\min_c q_c(\mathbf{k})$, and $q_c(\mathbf{k}) \geq \lambda_{\min} |\mathbf{k}|^2$ by Lemma 2. $\square$

Proposition 15 (Two-channel criterion). *Let $C = 2$ with isotropic tensors $\mathbf{D}_c = d_c I$, $\mathbf{J} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and write $\sigma = |\mathbf{k}|^2$. Then $\mu(\mathbf{k}) > 0$ for some $\sigma > 0$ while $\mu(0) < 0$ if and only if*

$$a + d < 0, \quad ad - bc > 0, \quad d_2 a + d_1 d > 2\sqrt{d_1 d_2 (ad - bc)}. \quad (12)$$

553 *Proof sketch.* M has trace $a + d - (d_1 + d_2)\sigma$ and determinant $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma +$
 554 $(ad - bc)$. The first two conditions give stability at $\sigma = 0$; the trace is then negative for all $\sigma > 0$, so
 555 instability requires $h(\sigma) < 0$ for some $\sigma > 0$. The minimum of the convex parabola h is attained at
 556 $\sigma^* = (d_2 a + d_1 d)/(2d_1 d_2)$ and is negative precisely under the third condition. $\square$

557 Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—
 558 the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Sec-
 559 tion 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreas-
 560 ing (Fig. 5c), and $\mu^*(k) < 0$ for all $k > 0$ in every run.

561 *Remark 16* (Why the negative result is expected). The training objective (3) rewards making the
 562 *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the
 563 homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated
 564 as a fixed point rather than as the outcome of an instability, the fitted operator is driven toward
 565 dissipativity. Detecting pattern-forming instability would require observations away from steady
 566 state, consistent with Corollary 12.

567 G Discretization invariance of the encoder

568 The encoder must map measurements on an arbitrary point set to a field. Let μ be the sampling
 569 measure on Ω and consider the kernel-integral operator

$$(\mathcal{K}v)(x) = \int_{\Omega} \kappa_{\theta}(y - x) \odot v(y) d\mu(y), \quad (13)$$

570 of which the graph layer is the empirical counterpart $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_{\theta}(x_j -$
 571 $x_i) \odot v(x_j)$.

572 **Proposition 17** (Monte-Carlo consistency). *If $\{x_j\}_{j=1}^N$ are i.i.d. draws from μ and $\kappa_{\theta} \odot v \in L^2(\mu)$,*
 573 *then for each x , $\mathbb{E}[(\mathcal{K}_N v)(x)] = (\mathcal{K}v)(x)$ and $\text{Var}[(\mathcal{K}_N v)(x)] = O(N^{-1})$; consequently $\|\mathcal{K}_N v -$*
 574 *$\mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$.*

575 *Proof.* The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbi-
 576 asedness and the $O(N^{-1})$ variance are immediate, and the norm bound follows by Chebyshev and
 577 Fubini. $\square$

578 Proposition 17 is the sense in which the encoder is discretization-invariant: the population operator
 579 (13) depends on the geometry through $\kappa_{\theta}(y - x)$ alone and not on the sampling pattern, and the
 580 empirical operator converges to it as the sampling density increases, whatever the lattice. **This**
 581 **is an asymptotic statement about the encoder, and it is worth emphasising that it does not**
 582 **imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that
 583 it does not: $\mathbf{D}_{\theta}$ and f_{θ} are fitted to one tissue’s stationary field, and Theorem 11 indicates they
 584 are unconstrained away from that field’s range. Discretization invariance of the *architecture* and
 585 portability of the *fit* are distinct properties, and only the former is established here.

586 **Proposition 18** (Splatting consistency). *Let $w_{\sigma}(r) = e^{-r^2/2\sigma^2}$ and define the Nadaraya–Watson*
 587 *field $F_{\sigma}(u) = \sum_i w_{\sigma}(\|u - x_i\|)v_i / \sum_i w_{\sigma}(\|u - x_i\|)$. If $v_i = v(x_i)$ for $v \in C^2$ and the sampling*
 588 *density p is bounded away from zero, then $F_{\sigma}(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$.*

589 *Proof sketch.* Standard kernel-regression bias–variance decomposition in two dimensions: the bias
 590 term is $\frac{\sigma^2}{2}(\Delta v + 2\nabla v \cdot \nabla \log p) + o(\sigma^2)$ and the variance is $O((N\sigma^2)^{-1})$. $\square$

591 The learnable bandwidth σ therefore trades a smoothing bias against sampling noise. Since the
 592 bias term is proportional to Δv , splatting contributes a further diffusive smoothing on top of the
 593 operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and
 594 one that the present work does not disentangle.

Complexity

Let N be the number of measured locations, G genes, C latent channels, $H \times W$ the lattice, E the number of graph edges and T the integration horizon. The costs per forward pass are: gene projection $O(NGd)$; graph layers $O(LEd)$; splatting $O(Nr^2C)$ for radius r ; spectral blocks $O(BCHW \log HW)$; operator $O(TCHW \log HW)$; decoding $O(NCd + NGd)$. The operator term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used here ($C=32$, $H=W=64$, $T=8$) it is under 4% of the forward cost—consistent with the parameter count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin over the strongest baseline.

H Empirical details and full result tables

This appendix collects the complete result tables underlying Section 5. Tables are placed inline with [h] so that each appears with its heading.

H.1 Single-section model comparison

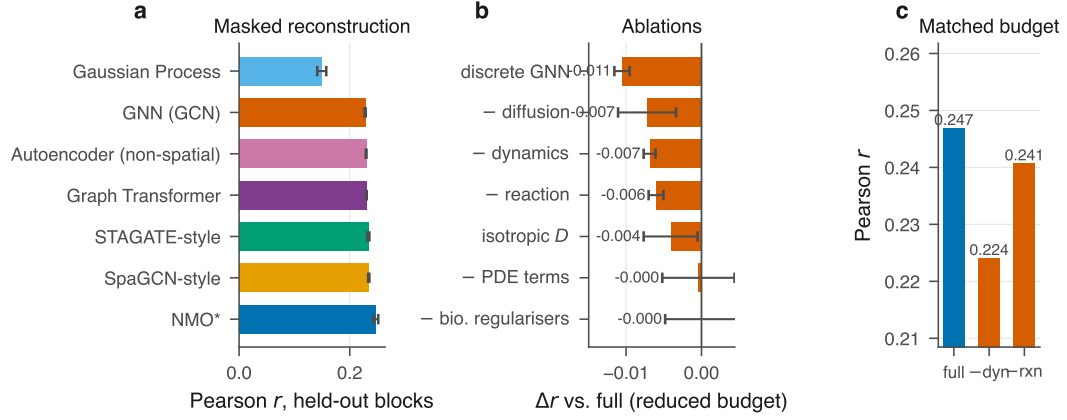


Figure 7: **Locating the effect** on the primary section, where all nine models are run. (a) Full model comparison. (b) Ablation deltas at the reduced budget. (c) The two decisive controls at the benchmark budget: disabling the operator ($T=0$) costs more than the full margin over the best baseline.

H.2 Numerical behavior

H.3 Paired statistics, numerics and biology

Table 3: Paired comparison of NMO against each baseline, with the tissue section as the unit of analysis. Δ is the mean paired difference in Pearson r (positive favors NMO) with a percentile bootstrap 95% CI over sections; d_z is Cohen’s effect size for paired designs; p is the Wilcoxon signed-rank test and p_{Holm} its Holm–Bonferroni correction across the baseline family.

baseline	n	Δr	95% CI	d_z	p	p_{Holm}	wins
Neural field (SIREN)	15	+0.0652	[+0.0502, +0.0812]	2.01	6.1e-05	0.00031	15/15
GP, multi-bandwidth	15	+0.0465	[+0.0374, +0.0559]	2.44	6.1e-05	0.00031	15/15
Autoencoder (non-spatial)	15	+0.0177	[+0.0095, +0.0257]	1.06	0.0015	0.0031	12/15
GNN (GCN)	15	+0.0166	[+0.0096, +0.0235]	1.16	0.00043	0.0013	14/15
STAGATE-style	15	+0.0165	[+0.0086, +0.0245]	1.00	0.002	0.0031	12/15

Table 4: Numerical behavior of the integrator variants. The largest stable step is measured by sweeping Δt and testing boundedness over 300 steps; the CFL reference is $h^2/(4\lambda_{\max}) = 0.01215$ for the explicit five-point Laplacian. Cost is the number of steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme. The empirical limit for `euler-fd5` tracks the CFL bound, confirming the analysis.

scheme	largest stable Δt	vs. CFL	steps to 10^{-2}	cost
strang-spectral	0.501	$41\times$	4	$1\times$
euler-spectral	0.00302	$0\times$	128	$32\times$
strang-fd5	0.0166	$1\times$	32	$8\times$
euler-fd5	0.0094	$1\times$	64	$16\times$

Table 5: Biological preservation of the reconstruction, averaged over 4 sections with reference annotations (Space Ranger clusters, Xenium clusters, and curated anatomical regions). Predicted expression at held-out locations is clustered and scored against the reference; ARI retention is the ratio to the score obtained from the measured data, so 1.0 would mean the reconstruction is as informative as the measurement. Marker AUROC asks whether the predicted field still discriminates a region using that region’s measured markers; k -NN preservation is the overlap of neighborhoods in measured versus predicted expression space; $|\Delta C|$ is the error in Geary’s C .

model	ARI $\uparrow$	ARI ret. $\uparrow$	NMI $\uparrow$	marker AUROC $\uparrow$	k -NN pres. $\uparrow$	$ \Delta C \downarrow$
GNN (GCN)	0.183 ± 0.065	0.495 ± 0.318	0.365 ± 0.071	0.764 ± 0.028	0.181 ± 0.052	0.727 ± 0.041
GP, multi-bandwidth	0.183 ± 0.045	0.408 ± 0.145	0.395 ± 0.041	0.775 ± 0.086	0.221 ± 0.059	0.727 ± 0.038
STAGATE-style	0.179 ± 0.041	0.467 ± 0.224	0.375 ± 0.058	0.768 ± 0.026	0.184 ± 0.053	0.715 ± 0.036
Neural field (SIREN)	0.091 ± 0.060	0.182 ± 0.110	0.219 ± 0.131	0.696 ± 0.058	0.106 ± 0.027	0.537 ± 0.062
NMO (ours)	0.197 ± 0.051	0.518 ± 0.276	0.407 ± 0.086	0.779 ± 0.027	0.228 ± 0.082	0.771 ± 0.051

Table 6: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_01	MERFISH, 500-gene pane	Mus	23,376	500	4178×4097	single cell
merfish_allen_05	MERFISH, 500-gene pane	Mus	45,896	501	6127×5166	single cell
merfish_allen_10	MERFISH, 500-gene pane	Mus	42,070	503	8402×5441	single cell
merfish_allen_14	MERFISH, 500-gene pane	Mus	94,357	500	9048×5717	single cell
merfish_allen_18	MERFISH, 500-gene pane	Mus	44,309	500	7980×4825	single cell
merfish_allen_26	MERFISH, 500-gene pane	Mus	79,282	501	9460×5685	single cell
merfish_allen_30	MERFISH, 500-gene pane	Mus	91,017	501	9870×5843	single cell
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837×6094	single cell
merfish_allen_36	MERFISH, 500-gene pane	Mus	115,675	503	10074×6559	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000×6961	single cell
merfish_allen_42	MERFISH, 500-gene pane	Mus	75,433	503	9875×7086	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806×6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500×4950	bin50 (25 μm)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975×2650	bin50 (25 μm)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_human_breast	Visium (55 μm spots, h	Homo	4,869	2,087	6395×6750	spot (multi-cell)
visium_mouse_brain	Visium (55 μm spots, l	Mus	2,691	2,079	4822×6379	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212×3128	single cell

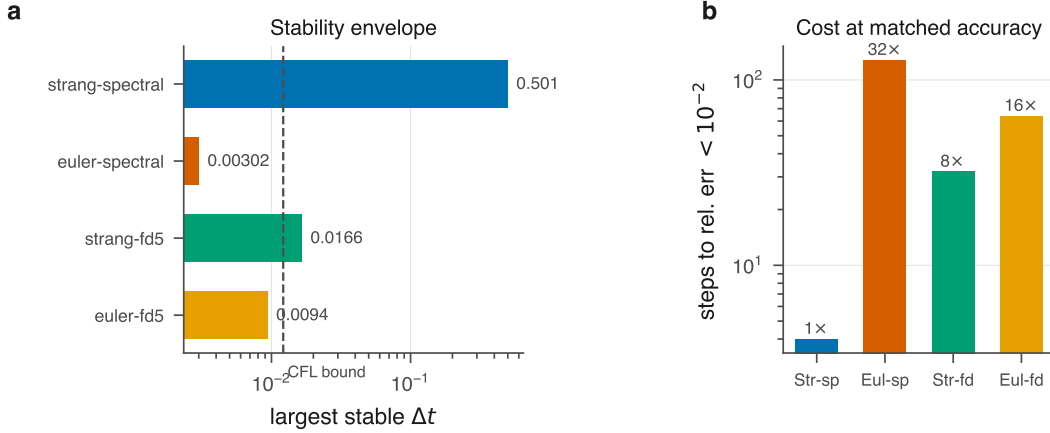


Figure 8: **What the exact spectral solve buys.** (a) Largest stable step size per scheme against the CFL bound for the explicit five-point Laplacian. (b) Steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme.

610 H.4 Dataset inventory

611 H.5 Ablation sweep

Full sweep at the reduced budget; the two matched-budget controls are reported in Section 6.

Table 7: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The *— dynamics* row is the critical control: it disables the operator entirely ($T=0$) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than the main benchmark** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	0.233 $\pm$ 0.005	—
— dynamics ($T=0$)	2.20M	0.227 $\pm$ 0.001	-0.007
— diffusion term	2.20M	0.226 $\pm$ 0.004	-0.007
— reaction term	2.19M	0.227 $\pm$ 0.001	-0.006
— PDE constraints	2.20M	0.233 $\pm$ 0.005	-0.000
— biological regularizers	2.20M	0.233 $\pm$ 0.005	-0.000
isotropic D	2.20M	0.229 $\pm$ 0.004	-0.004
state-dependent D	2.20M	0.236 $\pm$ 0.004	+0.003
discrete GNN operator	0.61M	0.223 $\pm$ 0.001	-0.011
latent $C=8$	1.07M	0.213 $\pm$ 0.005	-0.021
latent $C=16$	1.30M	0.220 $\pm$ 0.002	-0.013
latent $C=64$	5.77M	0.237 $\pm$ 0.000	+0.004

612

613 H.6 Cross-tissue and cross-resolution transfer

614 H.7 Developmental forecasting

615 Horizon sweep for E9.5 $\rightarrow$ E10.5. Consecutive MOSTA stages are distinct embryos, so the compar-
616 ison is at the level of the rasterized field after isotropic normalization to a common frame.

Table 8: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
STAGATE-style	0.212 ± 0.001	0.249 ± 0.000	0.358 ± 0.001
NMO (ours)	0.209 ± 0.004	0.251 ± 0.003	0.365 ± 0.004
<i>target (zero-shot)</i>			
STAGATE-style	0.093 ± 0.001	0.397 ± 0.002	0.266 ± 0.002
NMO (ours)	0.033 ± 0.012	0.409 ± 0.005	0.219 ± 0.017
<i>target (decoder-only fine-tune)</i>			
STAGATE-style	0.117 ± 0.007	0.393 ± 0.000	0.365 ± 0.003
NMO (ours)	0.041 ± 0.019	0.406 ± 0.004	0.282 ± 0.017
<i>target (in-domain oracle)</i>			
STAGATE-style	0.147 ± 0.001	0.375 ± 0.000	0.427 ± 0.001
NMO (ours)	0.136 ± 0.001	0.397 ± 0.002	0.329 ± 0.003
<i>target (training-mean floor)</i>			
mean	-0.000 ± 0.000	0.404 ± 0.002	0.254 ± 0.001

Table 9: Cross-resolution transfer: $55 \mu\text{m}$ Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NMO (ours)	0.256 ± 0.009	0.246 ± 0.002	0.384 ± 0.007
STAGATE-style	0.242 ± 0.001	0.248 ± 0.001	0.364 ± 0.001
<i>target (zero-shot)</i>			
STAGATE-style	0.282 ± 0.000	1.765 ± 0.001	0.208 ± 0.001
NMO (ours)	0.139 ± 0.015	1.863 ± 0.026	0.125 ± 0.004
<i>target (in-domain oracle)</i>			
NMO (ours)	0.259 ± 0.006	1.752 ± 0.007	0.556 ± 0.006
STAGATE-style	0.256 ± 0.001	1.760 ± 0.002	0.569 ± 0.001
<i>target (training-mean floor)</i>			
mean	0.000 ± 0.000	1.872 ± 0.000	0.101 ± 0.001

Table 10: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterized field after isotropic normalization to a common frame; it inherits the error of that coarse registration. *Persistence* predicts E10.5 = E9.5 and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NMO	$T=0$	0.082 ± 0.001	0.320	74%
NMO	$T=2$	0.089 ± 0.000	0.305	72%
NMO	$T=4$	0.092 ± 0.000	0.297	73%
NMO	$T=8$	0.094 ± 0.000	0.289	72%
NMO	$T=16$	0.095 ± 0.001	0.283	72%
NMO	$T=32$	0.090 ± 0.003	0.285	70%

617 H.8 In-silico morphogen source

618 Split-half protocol of Section 5.6: the perturbation direction is defined from a random half of each
619 pathway and enrichment scored on the held-out half, against a permutation null over matched gene sets.

Table 11: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 ± 6.0	49.7	0.475	575 ± 154
FGF	19	72.6 ± 0.8	50.3	0.992	618 ± 58
SHH	15	56.6 ± 6.7	50.4	0.698	466
WNT	26	56.4 ± 3.9	49.9	0.762	649 ± 143

620

621 H.9 Perturb-seq consistency

622 An out-of-context probe: Norman et al. profiled non-spatial K562 cells whereas the operator is fitted
623 to tissue. With $n = 7$ testable perturbations the comparison is underpowered and we report it as inconclusive.

Table 12: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signaling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NMO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

624

625 H.10 Latent fields

626 I Reproducibility

627 All datasets download automatically with a recorded SHA256 manifest; every table and figure in this
628 paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints
629 are released.

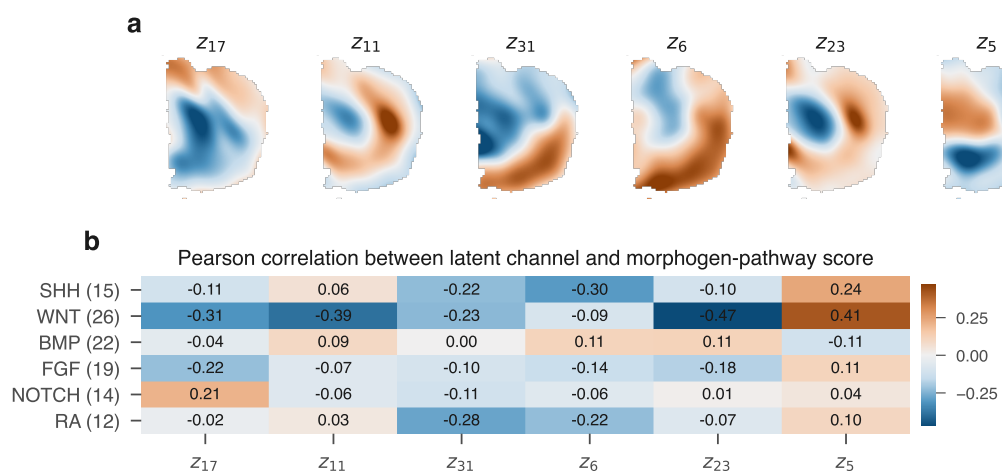


Figure 9: Latent channels of the relaxed field (a) and their correlation with canonical morphogen-pathway scores (b). Section 5.6 shows these correlations do not survive a causal split-half test.